Universidad del Desarrollo
Facultad de Ingeniería

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Instituto Data Science

W1 · Discovering questions worth investigating

Business Intelligence · IIB423T-1
Universidad del Desarrollo · Facultad de Ingeniería
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Assessment information	Details
Guided work	Wednesday 23 September 2026, 17:50–20:20; break 19:00–19:10
Submission deadline	Thursday **24 September 2026, 23:59**
Time zone	Santiago, Chile
Teams	Same team as C1; pairs preferred; authorized individual work remains available
Weight	**10% of the pre-exam course grade (NP)**
Points	**100 shared team points**; individual contributions documented
Tools	Python/Jupyter recommended; other verifiable tools accepted; GitHub optional

Your challenge: bring at least two credible lines of inquiry to the instructor, supported by documentary evidence and a first inspection of real data. Make the options concrete enough to receive useful feedback before choosing the line you will develop in C1.

1. The observatory and your role

You are contributing to an educational observatory of health data in Chile. Its value will come from several useful questions, not from every team answering a single predefined problem. Public information about health services and territories offers many possibilities, but a large collection of files is not yet an investigation.

Your task is to discover a relevant problem, examine what is known about it and determine what the available data can reveal. These are analytical decisions. If you only followed a recipe, you could produce correct code without learning how to choose a worthwhile question or recognize an invalid measurement.

AI can support exploration and execution, but you must verify its sources and suggestions. The assessment values the reasoning behind your choices and the evidence you inspect. More code, more datasets or a dramatic result will not automatically earn more points.

Use literature, technical reports and official health priorities to establish context. No interviews or external stakeholder review are required. You may propose who could use the analysis and for what decision; do not claim to know their views without evidence.

2. Expected learning outcomes

These are English renderings of the approved course outcomes. W1 provides initial evidence toward them; it does not assess complete mastery of the course.

Course learning outcome	What you will demonstrate in W1	Rubric
LO1. Analyze complex organizational problems and integrate information from different areas to formulate solutions.	Formulate and compare relevant candidate problems using documentary evidence and available data.	W1-A, W1-C
LO2. Use digital technologies and computational techniques to process, model and represent data in organizational contexts.	Demonstrate initial **inspection and representation** of real data, with traceable procedures. Modeling is not assessed in W1.	W1-B, W1-D
LO3. Interpret patterns, trends and relationships in data to support informed decisions.	Interpret initial observations cautiously and use them to judge the feasibility of the proposed lines.	W1-B, W1-C

W1 is the first checkpoint toward C1. You are not expected to complete two full EDA projects or commit to a final topic before receiving feedback.

3. What to investigate

A. Build at least two candidate lines

For each candidate, explain:

- **The problem:** what concern you would investigate in Chile, for which population or service, where and over what period.

- **Its relevance:** what literature or official technical evidence supports the concern. Explain any applicable connection to a MINSAL priority or plan, without forcing a match. Cite retrievable sources with institution/author, title, date and link.

- **A tentative question and use:** what you would ask, who could use the answer and what decision it could inform. Label the proposed user and use as your proposal.

- **The data opportunity:** which public sources might answer the question and what information could be missing.

The alternatives must represent meaningfully different questions or decision uses, not the same question with a renamed title. They may share a dataset. There is no requirement to use different tools or unrelated clinical topics. Two well-examined candidates are sufficient; additional candidates do not automatically earn points.

News may help you locate an issue, but follow important claims to research, official reports or technical documentation. A policy goal expresses what should improve; your analysis must establish separately whether the data can measure it. Record the year/version of any policy or numerical target.

B. Inspect actual data for each line

Open the relevant file or a clearly labeled subset. Use its dictionary to establish what one row represents, the important fields, period, coverage and possible identifiers. Show an initial summary, comparison or plot that helps assess the candidate, with a short interpretation and limitation. Identify relevant first quality checks using the Five Cs; full cleaning is not expected yet.

If a candidate proves infeasible before a meaningful summary is possible, document the concrete evidence: for example, a dictionary and field inspection showing the needed event is absent, or a verified access restriction. Explain why this blocks the question. A well-supported rejection can receive full credit; unsupported claims that the data are unavailable cannot replace exploration. At least one candidate must include actual data inspection with a relevant summary/comparison/plot, so the team has a workable starting point for C1.

Identify a plausible join if it is relevant, including the intended key and observation units; label it **proposed** until tested. You do not need to join every source in W1. There is no guaranteed universal key across the course datasets. A shared code alone does not establish compatible meanings, periods or populations.

Use the seven steps from Class 03 to notice gaps: decision, success criteria, business questions, business process, entities/events/states, data representation and measurement. W1 asks for an initial framing; C1 develops the selected line fully.

C. Compare your options and request feedback

Bring a compact comparison to the instructor. A table can help:

Dimension	Candidate A	Candidate B
Relevance supported by sources	Your evidence	Your evidence
Fit between question and inspected data	What can/cannot be observed	What can/cannot be observed
Analytical opportunity for C1	What you could investigate	What you could investigate
Main limitation or uncertainty	Its consequence	Its consequence
Feasible scope and future ML/dashboard potential	Possible task, users and data gaps	Possible task, users and data gaps

Explain your provisional preference and ask specific questions where instructor experience could change your decision. Do not invent a weak second candidate after making your choice. If you change direction during exploration, preserve that reasoning in the record.

You may maintain, adapt or reject advice with evidence. Agreement with the instructor is not a grading criterion. Presenting uncertainty clearly is stronger than pretending that a source can answer a question it cannot.

D. Project continuity: beyond C1

The project will continue toward **machine learning and a dashboard** later in the course. Consider this when comparing both candidates, even though neither implementation is required in W1 or C1.

For each candidate, propose a possible future ML question, such as predicting a meaningful outcome or classifying a relevant event. Identify what data might make it possible and what remains to be checked. State who could use a dashboard, for what decision and which indicators or comparisons might help them. A statement such as “we will apply AI later” is not a feasibility argument.

At this stage, uncertainties are expected: do not select algorithms, train models, design a complete validation protocol or promise predictive accuracy. A documented lack of a useful or feasible extension can justify rejecting a candidate. Use instructor feedback to identify a line that can support both the present EDA and a plausible later extension, without forcing ML into a problem where it adds no value.

4. Working and submitting

Use the two regular class blocks for investigation and instructor conversations. The 140 working minutes are available for progress; no additional recovery module is included. In a pair, each member may lead a candidate, but compare the options together. An authorized solo submission explores the same two candidates without inventing a collaborator.

One archive with a mandatory internal structure

Submit **one ZIP or RAR file** through the designated Canvas assignment. Use `GROUP_XX_W1.zip` or `GROUP_XX_W1.rar`, where `XX` is your two-digit group ID. For example: `GROUP_01_W1.rar`. Keep the same group ID in W1 and C1; use the actual archive format, not a renamed extension. In the generic pattern `GROUP_XX_ASSESSMENT`, replace `ASSESSMENT` with `W1` or `C1`.

The archive must contain a single root folder with the same name, without the extension. The following structure and the report filename are mandatory; replace `01` with your group ID. Choose descriptive filenames inside `analysis/` and `data/`.

```
GROUP_01_W1/
  README.md
  report/
    GROUP_01_W1_Report.pdf
  analysis/
    [editable notebooks, scripts, workbooks or queries]
  data/
    [reduced input data and generated case tables]
```

The report PDF is mandatory and self-contained. The report must contain both candidate briefs, their documentary support, first data observations, comparison, provisional preference, questions for feedback and future ML/dashboard potential, followed by the Process record and references. The instructor must be able to understand the complete submitted investigation from this PDF alone, without opening a notebook, workbook or the project. Explain the methods and consequential decisions, show the relevant tables/figures with units and context, and interpret the findings and limitations. “See the notebook” is not a substitute for the explanation or evidence needed to understand your argument. Technical files support verification and reproduction, not the basic reading of the report. Notebooks and workbooks must still document the purpose, procedures and interpretation of their analysis blocks; do not paste every code cell into the PDF.

Submit executed notebooks with saved outputs. Before submission, restart the kernel, run the submitted analysis from top to bottom in the documented order, and save the `.ipynb` with its tables, figures and relevant outputs visible. Outputs must correspond to the submitted code and data; do not clear them before uploading or mix stale results from an earlier version. If an unresolved error prevents a part from running, identify the affected section and its consequences in the PDF and README rather than hiding it; this does not waive the requirement for executed evidence. For Excel, Power Query or other tools, save the calculated/refreshed results, formulas and transformation steps so the reported evidence is inspectable without first rerunning everything.

`README.md` **is the package entry point.** Identify members and group ID, submission date/version, file roles and the order in which to open/run/refresh them. State the software and versions used, required libraries or connectors and how to install/configure the dependencies. For Python, a requirements file may complement these instructions. Explain paths, source versions and any unresolved execution issue. Use project-relative paths rather than paths tied to your own computer.

Include the project files and the data for your reduced case. In `data/`, include the relevant smaller inputs or an uncleaned extract for the chosen scope, as well as generated tables needed to verify the reported results. Label which files are extracted inputs and which are processed outputs; document their source, scope and transformations. In W1 this means the data actually inspected so far, not a completed cleaning pipeline for both alternatives. A candidate rejected before access should have its infeasibility evidence in the report.

You may omit giant original datasets, but the README must identify the exact public source or course file/version, retrieval instructions and the script/query or documented filters that reconstruct the reduced input. Where downloads can change, record the version/date used and retain your local original. The reviewer should be able to inspect the reduced case without downloading the entire collection and trace how that case was obtained. A final summary table alone is insufficient if its preparation cannot be checked.

Include the relevant editable files, saved outputs and available source documents used to prepare the delivery; do not include installed applications, Python installations, `.venv`/Conda environment folders, caches, temporary files or credentials. Describe dependencies instead. This is a project snapshot, not a copy of the entire computer. Check that the archive opens and that its internal paths work after extraction.

Process record: a required section of the report PDF

Place a section titled **Process record** after the analytical brief and before the references. It explains how the work developed; it is not a dump of code, chats or every command executed. Use the following subsections and concise tables where helpful:

Subsection	What to record
Decisions and checks	Date/stage; consequential decision; alternative considered; reason and supporting evidence; check performed; result or change; location in the submitted files.
Instructor feedback and response	Advice received and when; whether you accepted, adapted or rejected it; your reason; change made and evidence location. If feedback is pending, state that and your provisional decision.
Individual contributions	Member; concrete contribution; location of evidence. Record actual contributions, not only intended task allocation. Equal volume is not required.
AI use and verification	Tool and purpose; relevant prompt/output excerpts; what you accepted, modified or rejected; how you checked consequential suggestions. State explicitly if no AI was used.
Individual reflection	Each member identifies a contribution, a learning gain and one unresolved uncertainty. An authorized solo submission follows the same structure.

For example, a decision entry can follow: **Question/issue; alternatives; decision and reason; verification; evidence location**. Cite a notebook heading, workbook sheet/range, query or file path so the reviewer can inspect the evidence. Your PDF must contain the reasoning itself; links to notebooks or external chats cannot replace it. References, code comments and output explanations complement this record rather than substitute for it.

Python/Jupyter is recommended and used for course demonstrations. Excel, Power Query and other tools are acceptable if formulas, queries or procedures and results can be inspected. GitHub is optional; no repository URL or commit SHA is required.

AI is allowed during preparation. Record tool, purpose, relevant prompts/output excerpts and what you accepted, modified or rejected, with checks of consequential suggestions. If no AI was used, say so. Use public data and do not upload confidential information. You remain responsible for the submitted content and cited sources.

There is no separate mandatory W1 oral defense. The existing workshop late rule applies: up to 24 hours late incurs a deduction of **10 points out of 100 available**; after that window the submission closes, subject to formal institutional adjustments. The on-time deadline is 24 September at 23:59; the late window ends 25 September at 23:59. Absence does not create a separate makeup workshop.

5. Rubric and scoring

Choose one level per criterion: Fully achieved = 100% of its weight; Mostly achieved = 80%; Partially achieved = 40%; Not achieved = 0%. Use the fixed points below, without interpolation. All published points are whole numbers. Add them directly; no rounding is required. The instructor applies the institutional conversion to grades.

The **core evidence** under each criterion is necessary for Fully or Mostly achieved. Fully requires all described dimensions; Mostly permits only a minor omission that does not undermine the core evidence. Relevant but incomplete or materially flawed evidence is Partially achieved. Not achieved means no assessable demonstration of that criterion. The instructor resolves borderline cases with a descriptor-based explanation.

A justified rejection can satisfy the criteria through the evidence it provides. The same analytical defect receives no additional penalty outside this rubric; effects on separate criteria must be explained separately. Tool choice, data volume, number of charts or agreement with instructor advice do not determine points. An inability to inspect a submitted file is flagged for clarification, not automatically scored as missing work. Assisted review may support the instructor; final assessment decisions remain with the instructor.

W1-A · Relevance and documentary foundation — 25 points

Core evidence: at least two meaningfully distinct candidate problems with relevant, retrievable documentary support for each.

Level	Points	Descriptor
Fully achieved	25	Both candidates identify a bounded Chilean health concern, tentative question and proposed use; relevant documentary evidence supports each, and any policy connection is explained without equating a target with an observed result.
Mostly achieved	20	Both distinct problems have relevant, retrievable support and clear tentative questions. A contextual boundary or supporting policy detail is incomplete without changing their relevance.
Partially achieved	10	At least one relevant candidate or useful source is developed, but a second distinct line, support for one line or a sufficiently bounded question is missing or materially weak.
Not achieved	0	Provides neither an identifiable candidate health topic/problem nor relevant documentary evidence or reasoning for investigating one.

W1-B · Actual data inspection and initial understanding — 30 points

Core evidence: source-specific inspection for both lines or a documented infeasibility finding for one; at least one actual-data summary with interpretation; identifiable observation units and coverage for the inspected data.

Level	Points	Descriptor
Fully achieved	30	Examines relevant fields, units, periods and coverage for the candidates; provides interpretable initial outputs or a documented infeasibility finding; identifies relevant first quality checks and distinguishes proposed joins from verified ones.
Mostly achieved	24	Core evidence establishes what the inspected data represent and supports initial feasibility judgments. A secondary field detail or first quality check is incomplete, without invalidating those judgments.
Partially achieved	12	Shows some real inspection but covers only one line without evidence for the other, misidentifies a key unit/coverage feature, or provides outputs without a usable interpretation; also applies when both lines have documented infeasibility but neither has an actual-data summary.
Not achieved	0	Provides only proposed procedures or copied catalog descriptions, with no assessable inspection or concrete infeasibility evidence.

W1-C · Critical comparison and questions for feedback — 30 points

Core evidence: an evidence-based comparison of the alternatives, a provisional preference and a concrete uncertainty on which feedback is requested.

Level	Points	Descriptor
Fully achieved	30	Weighs relevance, question–data fit, analytical opportunity, limitations and scope; considers possible future ML tasks, dashboard uses and data gaps for both candidates or justifies rejection; explains the preference and what could change it; asks specific feedback questions.
Mostly achieved	24	Makes a supported comparison and provisional choice, acknowledges the main limitation, considers both ML and dashboard potential and requests specific feedback. A secondary trade-off or future-use/data detail is incomplete.
Partially achieved	12	Shows some comparison or reasoned preference, but omits the provisional preference, a concrete feedback question or consideration of ML/dashboard potential; alternatively, relies mainly on convenience or overlooks a decisive feasibility issue.
Not achieved	0	Provides neither a relevant comparison of the options nor reasoning for a provisional direction.

W1-D · Traceability, contributions and AI accountability — 15 points

Core evidence: identifiable sources and inspectable work, contribution evidence, and an explicit AI-use statement.

Level	Points	Descriptor
Fully achieved	15	The prescribed archive contains a report PDF understandable without the project, its Process record, executed/saved analysis with outputs, reduced-case data and usable README; sources, choices and contributions are traceable; reflections and relevant AI excerpts/checks are included.
Mostly achieved	12	The PDF explains the investigation on its own; Process record, saved analysis outputs and reduced data permit verification; contributions and AI use are explicit. A minor naming/navigation detail or nonessential process entry is incomplete.
Partially achieved	6	Provides some source or process trail, but the PDF/Process record is absent or the PDF depends on external files to explain the investigation; required analysis outputs, reduced inputs or editable work are missing; or reproduction, contributions or AI accountability have material gaps.
Not achieved	0	Provides no usable source or process trail for the submitted work.

6. Starting points and the next checkpoint

The course collection is organized into **Urgencias** (2020–2024), **REM**, **Establecimientos**, **GRD**, and **cartografia_censo_2024**, with the available dictionaries. These are options, not a checklist of datasets to use. More public sources may be added when suitable and documented.

- DEIS: health statistics, documentation and open data.

- MINSAL datasets on datos.gob.cl.

- FONASA GRD dashboard: inspect definitions and distinguish dashboard views from downloadable data.

- MINSAL National Health Strategy toward 2030, 2022 edition: a starting point for health priorities. Check applicable updates and indicator definitions for the chosen topic.

Use the instructor conversations during W1 and the **29 September** progress review to refine your direction. C1 is due **30 September before 17:50**, followed by the presentation and individual defense. C1 develops **one** selected line; it assesses your subsequent analysis and use of feedback, rather than awarding the W1 points a second time. If feedback has not yet arrived, document that fact and your provisional reasoning instead of inventing a response.